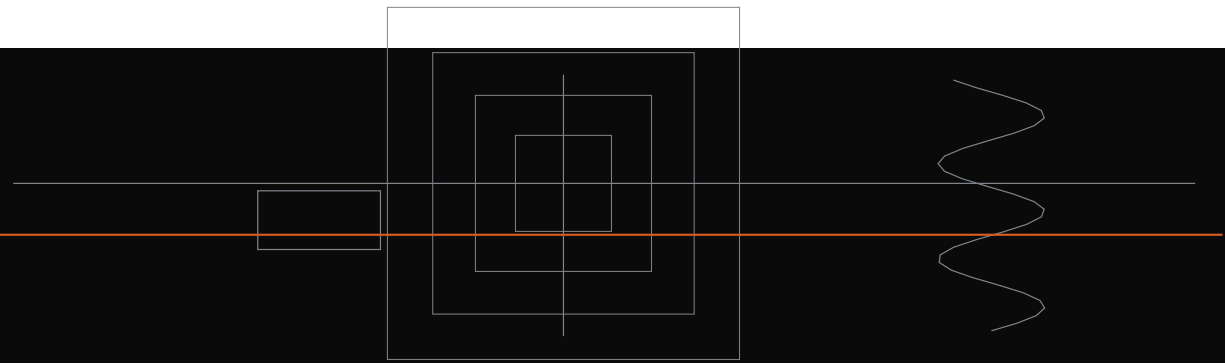


BILL OF MATERIALS

# Quadruped II

Rev. 2026-04-30 96 items



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**BOM**

Product/Project: Quadruped II  
Revision date: 2026-04-30

<https://www.printables.com/model/1451339-quadruped-ii>

**ELECTRICAL COMPONENTS**

22 items

|    | Description                     | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes                                                         |
|----|---------------------------------|-----|---------------------------------|-----------------------------|---------------------------------------------------------------|
| >> | 0001-1202 Power_Dist-PCB        | 1   | <a href="#">Order PCB HERE</a>  |                             | Power Distrivbution PCB - PCB Files available for Clubmembers |
| >> | Servo Motors                    | x12 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 180 degrees, 35kg (Approx 49.5x20,2x40mm)                     |
| -> | <small>ALT</small> Servo Motors | x12 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Budget friendly alternative.                                  |
| -> | <small>ALT</small> Servo Motors | x12 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Budget friendly alternative.                                  |
| >> | Raspberry Pi 4 Model B          | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 8gb Ram                                                       |
| >> | LiPo Battery                    | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 4000mAh 4S 45~90C Lipo Pack w/XT-90                           |
| >> | IMU                             | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Integrated Measuring Unit                                     |
| >> | Raspberry Pi Breakout Board     | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Made by Dfrobot                                               |
| >> | Fan                             | x2  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 60x60 12v                                                     |
| >> | Servo Driver PCA9685            | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Communication via I2C interface                               |
| >> | Arduino Nano                    | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For IMU, Battery sensing, LCD screen                          |
| >> | DC to DC (Buck) converter 20A   | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Adjustable, set to ~7V                                        |
| >> | DC to DC (buck) converter 3A    | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For fans                                                      |
| >> | 32GB MicroSD card               | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 32GB is maximum Pi supports                                   |
| >> | RasbPi Cooling System           | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -                                                             |
| >> | Arduino Nano Breakout Board     | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -                                                             |

**BOM**

Product/Project: Quadrupe II  
Revision date: 2026-04-30

<https://www.printables.com/model/1451339-quadrupe-ii>

**ELECTRICAL COMPONENTS continued**

|    | Description           | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes                                             |
|----|-----------------------|-----|---------------------------------|-----------------------------|---------------------------------------------------|
| >> | GPIO Cable            | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | To connect Pi to its breakout board               |
| >> | Logic Level Converter | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For adjusting voltage between the pi and the nano |
| >> | XT60H Connector       | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | To connect LiPo to the robot                      |
| >> | LCD Screen            | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 1.47 inch                                         |
| >> | PS4 Controller        | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -                                                 |
| >> | Digital Voltmeter     | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | DC 2.5-30V, used to measure battery voltage       |

**BOM**

Product/Project: Quadruped II  
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**MECHANICAL COMPONENTS**

28 items

|    | Description              | Qty        | AliExpress                      | Amazon (Geniuslink)         | Notes                                                           |
|----|--------------------------|------------|---------------------------------|-----------------------------|-----------------------------------------------------------------|
| >> | Bearing                  | <b>x4</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 6705-2RS 25X32X4                                                |
| >> | Thrust Bearings          | <b>x4</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | F4-10M 4x10x4mm                                                 |
| >> | Bearing                  | <b>x20</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 684ZZ Mini Deep Groove Ball Bearing 4x9x4mm                     |
| >> | Servo horns              | <b>x4</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | RC Car Steering Servo Horn Arm 25T Teeth Long                   |
| >> | Heatsinks                | <b>x12</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 6mm, black. Cut in half for each servo                          |
| >> | Servo Discs              | <b>x10</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | with screws.                                                    |
| >> | Threaded rods - Long     | <b>x4</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 x 110mm (cut to length if needed)                            |
| >> | Threaded rods - Short    | <b>x4</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 x 25mm (cut to length if needed)                             |
| >> | Magnets                  | <b>x8</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Used to hold fans in place                                      |
| >> | Ballt Joint (Long joint) | <b>x8</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | S - 3mm L30mm M3                                                |
| >> | Ball Join (Short Joint)  | <b>x8</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | N - 3mm L20mm M3                                                |
| >> | On/off Switch            | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 20A at 14VDC rated                                              |
| >> | Spring                   | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For battery latch mechanism. 35mm long x 7mm OD                 |
| >> | Smooth Rod 3x80          | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For battery hinge mechanism 3x75-80mm (cut to length if needed) |
| >> | Misc                     | <b>1</b>   | -                               | -                           | -                                                               |

## MECHANICAL COMPONENTS continued

|    | Description                        | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes                                                                               |
|----|------------------------------------|-----|---------------------------------|-----------------------------|-------------------------------------------------------------------------------------|
| >> | Male Connector (for peripherals)   | x12 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | B2B-PH-K-S(LF)(SN)                                                                  |
| >> | Socket for Connectors              | x25 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | SPH-002T-P0.5S                                                                      |
| >> | Female Connector (for peripherals) | x12 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | PHR-2                                                                               |
| >> | Capacitor                          | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 2200uF 16VDC Electrolytic RB Capacitor ,For servo board                             |
| >> | Reverse Polarity Protection Diode  | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | B160-13-F                                                                           |
| >> | Ferrules                           | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Optional, if used need approx 20 across robot                                       |
| >> | Terminal Blocks                    | x8  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 1935776                                                                             |
| >> | 100k resistors                     | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For dividing the voltage at the buck converter so that it is readable by the nano   |
| >> | 47k Resistor                       | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | CRGCQ1206F47K                                                                       |
| >> | 10k resistor                       | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | CRGCQ0805F10K                                                                       |
| >> | Heat Shrink                        | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Only small amount required, for covering exposed solid-core wires                   |
| >> | Power Delivery Wire                | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Minimum 12AWG, for wires expected to carry current above 2A and/or voltage above 5V |
| >> | Connection Wire                    | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 19-30AWG, recommend solid core, used for all other wires                            |

**BOM**

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**FASTENERS**

22 items

|    | Description         | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes                                  |
|----|---------------------|-----|---------------------------------|-----------------------------|----------------------------------------|
| >> | MC6S-M2x8           | x8  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M2 - Socket Head Cap Screw             |
| >> | M6CS-M2x4           | x14 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M2 - Socket Head Cap Screw             |
| >> | MC6S-M2.5x6         | x8  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M2.5 - Socket Head Cap Screw           |
| >> | MC6S-M3x8H          | x38 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Socket Head Cap Screw             |
| >> | MC6S-M3x10H         | x4  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Socket Head Cap Screw             |
| >> | MC6S-M3x14H         | x6  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Socket Head Cap Screw             |
| >> | MC6S-M3x20H         | x16 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Socket Head Cap Screw             |
| >> | MC6S-M3x25          | x4  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Socket Head Cap Screw             |
| >> | MC6S-M4x25H         | x4  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M4 - Socket Head Cap Screw             |
| >> | MC6S-M4x16H         | x4  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M4 - Socket Head Cap Screw             |
| >> | MF6S-M3x8           | x52 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Countersunk Socket Head Cap Screw |
| >> | K6S - M3x8          | x4  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Socket Button Head Screw          |
| >> | K6S - M3x10         | x10 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Socket Button Head Screw          |
| >> | K6S - M3x16         | x36 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Socket Button Head Screw          |
| >> | K6S - M3x20         | x20 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 - Socket Button Head Screw          |
| >> | RBS M3 (3,2x9x0,8)) | x4  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 Washer with Large O.D               |
| >> | LM6M_A4-70(M3)      | x24 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 Lock Nut                            |
| >> | ML6M M3 Nut         | x8  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 Nut - Low profile                   |

|            |                                                            |                                                                                                                           |
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|------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|

| FASTENERS continued |                        |            |                                 |                             |                  |
|---------------------|------------------------|------------|---------------------------------|-----------------------------|------------------|
|                     | Description            | Qty        | AliExpress                      | Amazon (Geniuslink)         | Notes            |
| >>                  | Heat insert M3 3x5     | <b>x96</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | <i>L=3 O.D=5</i> |
| >>                  | Heat Insert M2 3x3     | <b>x18</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | <i>L=3 O.D=3</i> |
| >>                  | Heat Insert M2.5 3x3.5 | <b>x8</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | <i>L=3 O.D=3</i> |
| >>                  | Heat Insert M4 4x6     | <b>x8</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | <i>L=4 O.D=6</i> |

|            |                                                            |                                                                                                                           |
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|------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|

| <div> <b>ASSORTMENT BOXES</b> <span>7 items</span> </div> |                                   |          |                                 |                             |                                  |
|-----------------------------------------------------------|-----------------------------------|----------|---------------------------------|-----------------------------|----------------------------------|
|                                                           | Description                       | Qty      | AliExpress                      | Amazon (Geniuslink)         | Notes                            |
| >>                                                        | Socket Button Head Screw          | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M2-M3 x 2510pcs                  |
| >>                                                        | Socket head cap Screw             | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 2500pcs M2M2.5M3                 |
| >>                                                        | Socket head cap Screw             | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 1410pcs M3M4M5                   |
| >>                                                        | Countersunk Socket Head Cap Screw | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 3045pcs - M2, M3, M4, M5, M6, M8 |
| >>                                                        | Heat Insert (with press)          | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 1040pcs - M2-M4                  |
| >>                                                        | Heat Inserts                      | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 1400pcs - M2-M4                  |
| >>                                                        | Lock Nut                          | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 230pcs M3,M4,M5,M6,M8            |



|            |                                                            |                                                                                                                           |
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|------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|

| <div>FILAMENT</div> <div>4 items</div> |                                                              |     |                                 |                             |       |
|----------------------------------------|--------------------------------------------------------------|-----|---------------------------------|-----------------------------|-------|
|                                        | Description                                                  | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes |
| >>                                     | PC / ASA / ABS / PETG Filament                               | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >>                                     | PC-CF / PETG-CF / PA-CF / ABS+ (Engineering Grade Filaments) | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >>                                     | PLA                                                          | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >>                                     | TPU                                                          | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |

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**TOOLS**

13 items

|    | Description                                 | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes                       |
|----|---------------------------------------------|-----|---------------------------------|-----------------------------|-----------------------------|
| >> | Soldering Iron TS101                        | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| >> | Heat Inset Tip Kit - For TS101              | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| >> | Electric Precision Screwdriver              | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| >> | Allen Key Set                               | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| >> | Torx Key Set                                | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| >> | Torque Screwdriver                          | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| >> | Wire stripper                               | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| >> | Wire cutter                                 | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| >> | Heat Insert Press Tool                      | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| >> | Crimp Tool                                  | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | (Recommended)               |
| -> | <small>ALT</small> Soldering Iron - Generic | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Budget friendly alternative |
| -> | <small>ALT</small> Heat insert tip - set    | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Budget friendly alternative |
| >> | Precision Screwdriver Set                   | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Budget friendly alternative |